

17

A

For wire X,

$$n = \frac{I}{vAq} = \frac{3.0}{(2.8 \times 10^{-2}) \times \pi (5.0 \times 10^{-5})^2 \times (1.6 \times 10^{-19})} = 8.526 \times 10^{28} \text{ m}^{-3}$$

n remains the same since wire X and Y are made from the same material.

For wire Y,

$$v = \frac{I}{nAq} = \frac{2.0}{(8.526 \times 10^{28}) \times \pi (1.0 \times 10^{-4})^2 \times (1.6 \times 10^{-19})} = 4.7 \times 10^{-3} \text{ m s}^{-1}$$

4